MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG4DSCECT203 Wireless Technology (Minor for Others)

Duration: 1½ hours **Maximum Marks: 50**

Students should attempt at least one question from each course outcome to enhance their overall *outcome attainability.*

**Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)*

Part A

Multiple Choice Questions

Answer **All** Questions

Each question carries 1 mark

1	The main purpose of modulation in wireless communication is to A. Reduce noise B. Increase antenna size C. Enable efficient transmission of signals D. Improve receiver sensitivity	U	1
2	Which frequency band is commonly used for unlicensed wireless communication? A. VHF B. ISM C. X band D. Ku band	K	1
3	The regulatory authority responsible for spectrum management in India is A. FCC B. ITU C. IEEE D. TRAI	K	1
4	In AM modulation, information is carried by variation in A. Frequency B. Phase C. Amplitude D. Pulse width	U	1
5	QPSK differs from BPSK by A. Using higher power B. Transmitting two bits per symbol C. Reducing bandwidth D. Increasing noise	U	1
6	Path loss in wireless communication increases mainly with A. Data rate B. Modulation scheme C. Distance D. Antenna height	U	2
7	The unit dBm represents power relative to A. 1 W B. 1 V C. 1 mW D. 1 μW	K	2
8	RSSI is used to measure A. Bandwidth	U	2

	B. Noise level C. Received signal strength D. Data rate		
9	The primary function of a mixer in an RF receiver is to A. Amplify signals B. Filter noise C. Convert frequencies D. Generate carriers	U	3
10	GSM, LTE and 5G are examples of A. RF components B. Modulation techniques C. Wireless standards D. Antenna types	K	3

1 * 10= 10

Part B

Short Answer Questions
Answer any **4** Questions
Each question carries 5 marks

11	Explain the basic concepts and applications of RF systems in wireless communication.	U	1
12	Describe RF spectrum allocation and explain the need for regulation in wireless communication.	U	1
13	Explain analog modulation techniques AM, FM and PM.	U	1
14	Explain path loss and fading in wireless communication channels.	U	2
15	Define SNR. Explain its significance in digital communication systems.	U	2
16	Describe the basic functions of antennas, amplifiers and filters used in RF systems.	U	3

4 * 5 = 20

Part C

Essay Questions
Answer any **2** Questions
Each question carries 10 marks

17	a) Explain digital modulation techniques ASK, FSK, PSK and QPSK. b) Discuss constellation diagrams and bandwidth versus data rate trade-off.	An	1
18	a) Explain link budget basics in wireless communication. b) Discuss the role of RSSI in signal quality assessment.	An	2
19	a) Explain RF oscillators and mixers with neat block diagrams. b) Describe their role in RF transmitter and receiver systems.	An	3
20	a) Explain the architecture and working principles of modern wireless communication systems such as Wi-Fi and Cellular Networks. b) Compare their applications.	An	2

2 * 10= 20